Master vehicle diagnostic guide

Shop reference for this truck. It separates three problems that can be present at the same time. A working theory is not a completed diagnosis. Every claim below carries a label.

CONFIRMED Design fact, standard code title, or a quoted factory bulletin. **CATALOG LISTED** Listed by the maker for this model. Still verify by VIN. **LIKELY** Fits the engine and the published layout. Confirm on the truck. **POSSIBLE** A credible cause. Not proven on this vehicle. **TEST REQUIRED** Do not replace a part until this measurement is made. **NOT CONFIRMED** Number or location that did not check out. Do not buy from it.

Vehicle identification

Item	What to use	Label
Vehicle	2004 Jeep Grand Cherokee, WJ, Laredo, as stated for this job. Read the VIN on the dash and the door sticker before ordering parts.	STATED
Engine	4.0L inline-six, 242 cu in. Chrysler sales code ERH. The 8th VIN character is usually S. Read it. Do not assume from the badge.	LIKELY
Air metering	Speed-density. The PCM uses the MAP sensor, RPM, and intake-air temperature. There is no mass-airflow sensor. A vacuum leak is not automatically "unmetered air" the way it is on a MAF engine.	CONFIRMED
Throttle and idle	Cable throttle. A 4-wire stepper IAC on the throttle body bypasses the closed plate. This is not drive-by-wire. Do not use an electronic-throttle relearn procedure.	CONFIRMED
Ignition	Coil rail, PCM-controlled. No distributor. Timing is not adjusted with a wrench.	CONFIRMED
Emissions	Described as the California package: four oxygen sensors, two mini-catalysts, and a main catalyst. Federal 4.0L trucks of this year use two sensors. Count the sensors and the mini-cats before you buy anything. Where the truck was purchased does not decide the package.	TEST REQUIRED
Transmission	Not assumed. Idle in Drive is a different load than idle in Park. Write down the range for every RPM reading.	TEST REQUIRED

Stop driving if either is true

The exhaust manifold is glowing orange or red, or engine oil is reaching the hot manifold, a downpipe, or an oxygen sensor. Cool the engine completely. Guides 05 and 06. Do not idle it "to see."

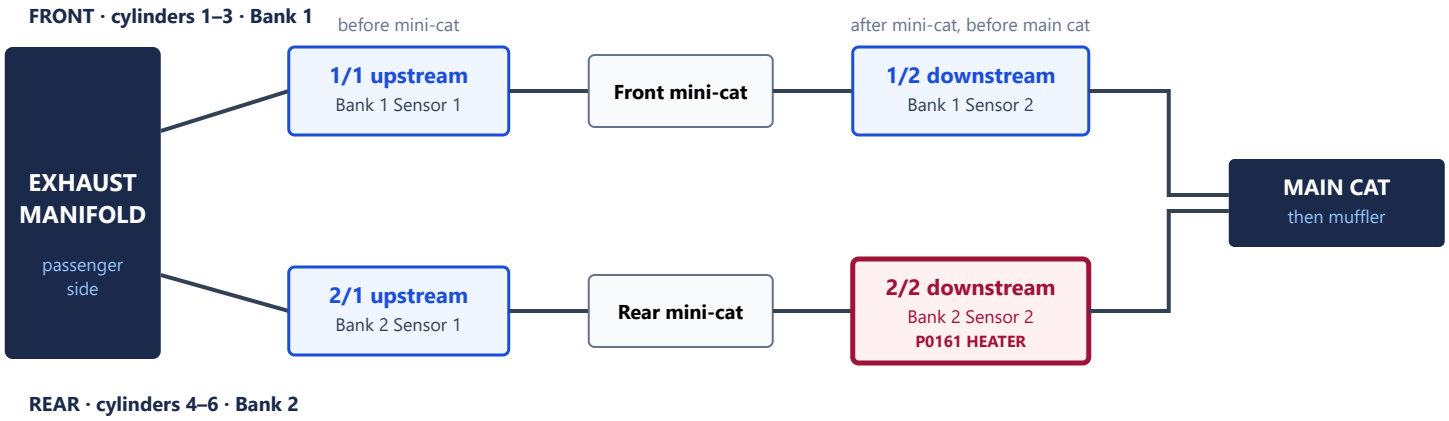
Print and use in this order

One-page sheet on the vehicle. Priority sheet next. Then the guide for the fault you are testing. Parts sheet last, and only after a test points at a part.

MASTER VEHICLE DIAGNOSTIC GUIDE · SENSOR LAYOUT

Four-sensor California layout

Published WJ layout for the 4.0L California package. **LIKELY** Confirm every bung on this truck before you unplug a sensor. Front of the engine is the radiator end. Cylinder 1 is at the front.



How the banks are named

Sensor 1 is upstream of that branch's mini-cat. Sensor 2 is downstream of the mini-cat and still ahead of the main cat. Bank 1 follows the front three cylinders. Bank 2 follows the rear three. **LIKELY** The harness length and the connector are different at each bung. Do not swap sensors as a test.

P0161 is not "the rear manifold sensor"

It is the heater circuit of Bank 2 Sensor 2: the rear downstream sensor, after the rear mini-cat. The published guide puts these sensors in the downpipes, not in the cast manifold. Trace the rear pipe.

TEST REQUIRED

Source of the four-position description: compiled WJ oxygen-sensor guide (JeepSpecs), consistent with repeated service descriptions of two mini-cats ahead of a main cat. It is not a page from the factory manual in hand. If this truck has only two sensors, stop and treat it as the federal package.

P0161 and P0505

P0161

CONFIRMED OBD-II title: O2 sensor heater circuit, Bank 2 Sensor 2.

The monitor is about the heater circuit. It is not a measurement of mixture, catalyst efficiency, or manifold temperature.

The circuit can fail because of: an open or shorted heater element, a corroded or oil-filled connector, a broken or melted wire, a fuse or ASD-relay feed, or the PCM control that grounds the heater. **POSSIBLE** each one. **TEST REQUIRED** before a sensor is bought.

This code does not prove the sensing element is dead, the engine is lean, or this sensor made the manifold glow. Oil on a connector can cause it. Oil on the sensor body more often ruins the signal than the heater. Test, then decide.

P0505

CONFIRMED Generic OBD-II title: idle control system malfunction.

On this Chrysler family the same code is often described as an **IAC motor circuit** fault: a short to ground or to battery on one of the four driver circuits while the motor is being commanded. That description comes from Chrysler-oriented service explanations, not from a 2004 FSM page photographed for this package. **LIKELY** as the electrical meaning. Read the wording on a Chrysler-capable scan tool.

A high idle is a separate observation. It may be stored beside this code. A related description, P1294, is “target idle not reached” on many Chrysler calibrations. If the tool offers it, record it. **TEST REQUIRED**

P0505 does not prove a vacuum leak. A leak can still be the reason the idle is high. Check both the circuit and the air leaks. Guide 03 and Guide 04.

Normal warm idle

CONFIRMED Chrysler TSB 18-008-04, domestic 2004 Grand Cherokee, 4.0L sales code ERH, built before 6 March 2004: the revised PCM software sets warm idle at **about 744 rpm**. That bulletin does not prove this VIN already has that software.

Screening band for this job: about 700–800 rpm, engine fully warm, Park, parking brake set, A/C and other loads off, after the idle relearn. Use a scan-tool RPM if you have one. In Drive, with the brake held, expect the PCM to hold a similar controlled idle, not a flare. Above about 1,000 rpm with no load, after a real relearn, is not a normal result. **TEST REQUIRED** Write down coolant temperature, gear, and whether the A/C is on. A cold engine idles higher on purpose. Do not judge the IAC until the engine is warm.

TSB source text: the new software “sets the engine idle speed at approximately 744 rpm (engine fully warm).” Subject of that bulletin was a low-idle drone and, on some Selec-Trac trucks, a transfer-case lamp. It is an idle-speed reference, not a diagnosis of this truck.

Symptoms and what they do not prove

What you see	Credible causes	Not proven by that symptom alone
Warm idle high or hunting	IAC pintle too far open, IAC O-ring leak, vacuum leak, throttle plate not closed, cable, carbon in the bore. POSSIBLE	A lean mixture. A glowing manifold. A failed O2 sensor.
P0505	IAC winding, connector, driver wire, or PCM driver. A pintle that was pushed can damage the motor electrically or mechanically. POSSIBLE	A vacuum leak. The leak can still be there. Ohms that look normal do not clear a pushed pintle.
P0161	Heater element, fuse, ASD feed, wiring, connector, PCM control. POSSIBLE	That the sensor signal is bad, or that this code caused a glow or a high idle.
Manifold glowing	Lean burn, misfire lighting fuel in the exhaust, retarded timing, restricted catalyst. POSSIBLE	That the IAC is the cause. Stop driving first. Guide 05.
Oil on the exhaust	Valve cover, oil-filter adapter, pressure sender, pan, rear main. POSSIBLE Trace fresh oil.	That the O2 heater failed. Fix the leak before a new sensor goes in.
Brown or purple manifold, seen cold	The iron has been very hot at some time. POSSIBLE past overheat.	That it is glowing now. An active glow is bright orange or red while hot. Stain alone is a clue, not a stop-driving event by itself.

Three problems. Do not collapse them into one.

Working theory POSSIBLE

Pintle was moved by hand → IAC may be damaged or held open → extra air → high or unstable idle → P0505 may set → mixture may go lean → exhaust may get hot enough to glow the manifold. Each arrow is unproven until the test beside it is done.

Separate hazard

Oil leak → oil on hot iron or on an O2 sensor → fire hazard. This chain does not depend on the IAC. It can destroy a new sensor and it can ignite if the manifold is hot. Guide 06.

Separate circuit

P0161 → Bank 2 Sensor 2 heater circuit → test connector, fuse, heater ohms, 12 V feed, and PCM control. Do not "fix" this code by replacing the IAC, and do not clear it and hope.

Why the lean step is only possible. This PCM is speed-density. Extra air from a manifold leak raises the MAP reading, and the PCM adds fuel for that reading. Idle still climbs, because the air is real. The mixture does not have to go lean. Prove a lean condition with upstream O2 voltage and fuel trims before you call the glow a lean-burn event. A misfire that dumps raw fuel into the pipe is an equally credible way to make iron glow.

MASTER VEHICLE DIAGNOSTIC GUIDE · DECISION PATH

Diagnostic decision tree

Short form. The full-page trees are Guide 08. Follow Guide 07 for the repair order. Do not start a road test while the manifold glows or oil is on the exhaust.

Start here — safety

Glowing manifold or oil on hot exhaust?
YES → stop, cool, Guides 05 and 06. NO → continue.

P0505, high idle, or a moved IAC pintle?

Measure warm idle. Screen 700–800 rpm in Park, loads off.
TSB reference about 744 rpm. TEST REQUIRED

High or unstable → smoke-test and inspect vacuum, CCV, EVAP, IAC O-ring, throttle plate.
Guide 04. Fix leaks before the relearn.

Pintle was pushed, or coils are open or shorted → new IAC. Do not touch the new pintle.
Guide 03. Ohms in range do not save a pushed motor.

Clean the throttle body. Plate must close. Cable must return.

PCM adaptive reset and idle relearn. Verify warm idle. Rescan.

P0161 — separate path

Identify Bank 2 Sensor 2 on the rear pipe, after the rear mini-cat.
TEST REQUIRED Count sensors first.

Connector, harness routing, oil in the plug, melted insulation.

Fuse and ASD feed. Use the under-hood lid diagram. Do not guess the cavity.

Heater resistance, sensor cold.
Screen about 4–10 Ω. Open = failed heater. TEST REQUIRED

About 10–12 V on the feed while the ASD is actually on, and PCM control on the other heater wire.

Repair the wire, fuse, or connector — or replace the sensor if the heater itself is open or shorted.

If oil is washing this sensor, fix the leak first or the new part dies the same way.

MASTER VEHICLE DIAGNOSTIC GUIDE · TEST RECORD

Test record — fill this in

Write the reading, not a conclusion, in the middle column. Judge it only after the guide for that test is followed. VIN last 8: _____ Mileage: _____ Coolant temp at idle check: _____

Test	Reading	Judgment / note
Sensor count (2 or 4) and mini-cat count		
Active manifold glow, or stain only, cold		
Oil reaching exhaust or a sensor? Fresh path from where?		
Warm RPM, Park, loads off		Screen 700–800. TSB ref. ~744
Warm RPM, Drive, brake held, loads off		Only if the vehicle is safe to put in gear
IAC coil A ohms / coil B ohms		Screen 40–80 Ω each. Isolated from each other
Was the pintle pushed or pulled by hand?		If yes, replace. Do not reuse.
Vacuum at warm idle, inHg, and does it flicker?		Typical screen about 17–21 inHg near this elevation
Smoke test: where smoke escaped		
Upstream fuel trims at idle and at 2,500 rpm		Record STFT and LTFT. Do not guess lean.
B2S2 heater ohms, cold, which cavities		Screen 4–10 Ω. OL = open heater
B2S2 heater volts, ASD on or engine running		Screen about 10–12 V
B2S2 control wire: does the PCM pull it?		Compare with another sensor if needed
Codes present before clearing, with freeze frame		
Codes after repair, idle verified		

Do not clear codes first. Freeze-frame RPM, coolant temperature, and fuel trims are the evidence. Photograph the scan tool. Clear only after the sheets above are filled and the repair that the tests support is done.

Elevation note: Tulsa is roughly 700 ft. A sea-level vacuum reading is a fair screen here. A low reading has more than one cause (leak, late timing, low compression, restricted exhaust). It is not a verdict.